

Shiny Numerati

My Community Dashboard v2

Jo-fai (Joe) Chow

Numerai Prague Community Meetup
2023-07-08



My Models: **JOE_THE_***



About Jo-fai (Joe) Chow

Background

Civil Engineer (Water Infrastructure)
Taught myself R & Python back in 2012

Work

Data Scientist / Meetup Organizer
(ex- Domino Data Lab & H2O.ai)

Contact

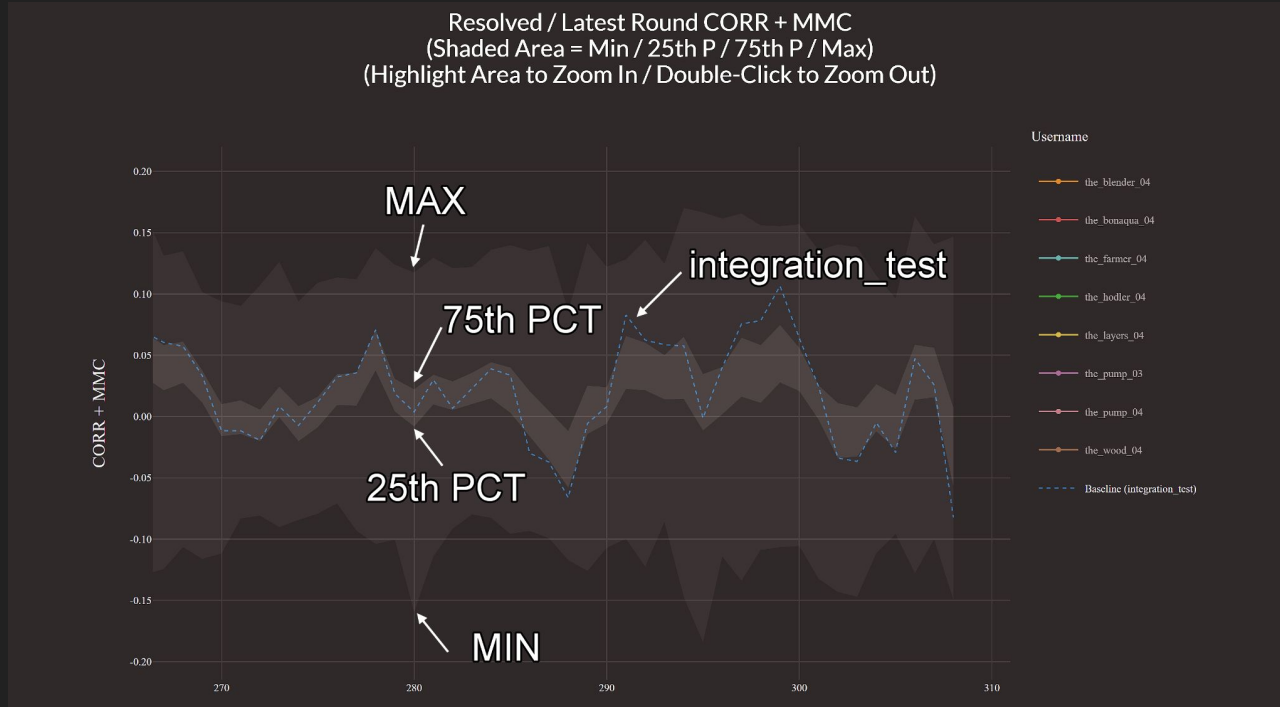
Twitter: [@matlabulous](https://twitter.com/matlabulous)
www.linkedin.com/in/jofaichow/

About this Talk

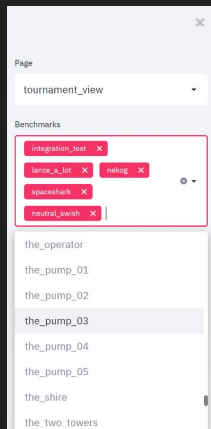
- A Brief History of My Numerati Dashboard(s)
 - Version 1
 - Version 1 + Colab with Ceunen
 - Version 2 - Shiny Numerati
- Dashboard Use Cases
 - Quick Overview of Numerai Benchmark Models
 - Building Mainstream vs. Anti-mainstream Models

A Brief History of My Dashboards

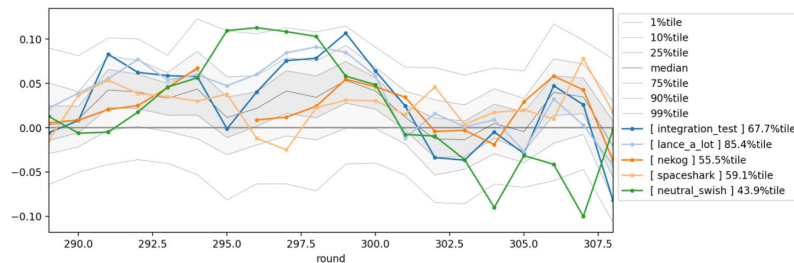
First version (2020 - early 2022): Numerati Dashboard



First Community Colab: Numerati + BC's Streamlit



Tournament correlations/mmc



	COZIMMC
lance_a_lot	0.0357
integration_test	0.0269
spaceshark	0.0235
nekog	0.0220
neutral_swish	0.0171



Bouwe Ceunen

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PAYOUTS

[dashboard.numeraipayouts.com](#)

Problems with the First Version of Numerati Dashboard

- No. of models: 300 → 10000+ 
- Standalone HTML: 1MB → 30 MB (GitHub 🙅)
- Key metrics: MMC → TC
- ... so I stopped maintaining the project in early 2022

My Second Attempt: Shiny Numerati

huggingface.co/spaces/jofaichow/shiny-numerati

The screenshot shows the 'Shiny Numerati' web application. On the left is a sidebar with navigation links: 'Start Here', 'Payout Summary', 'Model Performance', 'Raw Data', 'Community', and 'About'. The main content area is titled 'Shiny Numerati' and 'Community Dashboard for the Numerai Classic Tournament'. It is divided into two sections: 'Step 1 - Select Your Models' and 'Step 2 - Download Data'. In Step 1, a dropdown menu shows '12 models selected (out of 14889)'. Below it, a search box contains 'lgbm_v4'. A table lists various models, with 'lgbm_v4_jerome20' selected. In Step 2, a button labeled 'Download Data from Numerai' is visible. The footer indicates 'Powered by ❤️, 📊, Shiny, and 🏠 Spaces' and 'Version 0.1.3'.

A Twitter post from user Jo-fai (Joe) Chow (@matlabulous). The text reads: 'New feature in ✨ #Shiny Numerati: Payout Summary - compare any @numerai model payouts 📊'. Below this, it says 'Try it → huggingface.co/spaces/jofaichow/shiny-numerati'. At the bottom, it says 'Many thanks to @OmniAnalytics for the #Rnumerai #rstats 📊 github.com/OmniacsDAO/Rnumerai'. The post has three dots in the top right corner.



Quick Demo →

<https://jofaichow-shiny-numerati.hf.space/>

My Use Cases

Quick Overview of Benchmark Models (474 to 519)



- Goal #1: I don't want my models to burn like that during difficult times.
- Goal #2: ... but I still want to earn NMR like the benchmark models during good times.
- Constraint: I am a degen. I love $1x\text{CORR} + 3x\text{TC}$.
#YOLO #NotFinancialAdvice

- Solution (current):
 - Two Groups of Models
 - Mainstream Models
 - Anti-mainstream Models
 - Stake evenly on all models (this may change after Bor's talk)

Mainstream Models

- Tree-based / NN
- Feature selection
- Parameters tuning
- Feature neutralization
- Optimization: good CorrV2
Sharpe / Sortino Ratio
- ... in other words, it is just a pretty standard approach

Anti-mainstream Models

- Similar steps as mainstream
- **Additional step:** build my own benchmark models based on different targets (tree-based w/ same params as shown in example script)
- Optimization: good CorrV2 + **low correlation with benchmark models**

Reference - My Feature Selection Talk at NYC Meetup

Meetup #2 - 2022-09-24 - New York City

- Eventbrite - [Link](#)
- Welcoming Remarks by the Numerai Council of Elders - [Video](#)
- Talk #1 - A Complete Beginner's Guide to Numerai: From Zero to Hero by Alex Benton aka Aventurine - [Video](#) - [Slides](#)
- Talk #2 - Feature selection for the classic tournament by Jo-fai Chow aka ia_ai - [Video](#) - [Slides](#)
- Talk #3 - Numerai Signals by Suraj Parmar (Numerai Council of Elders) - [Video](#) - [Slides](#)
- Talk #4 - Multi-strategy hedge funds and their discontents, their past and their future by Richard Craib (Founder and CEO of Numerai) - [Video \(Pending Review\)](#)
- Talk #5 - Welcome to the AI Arena by Brandon Da Silva and Wei Xie (Co-founders of AI Arena) - [Video](#) - [Research Platform Walkthrough](#)
- Closing Remarks by the Numerai Council of Elders - [Video](#)

Observation #1:

My **mainstream** models still **burn** like the benchmark models 🥲



XGB
Mainstream
Models

NN
Mainstream
Models

Observation #2:

... but my **anti-mainstream** models can **survive** better 🙌



XGB
Anti-mainstream
Models

NN
Anti-mainstream
Models

Observation #3:

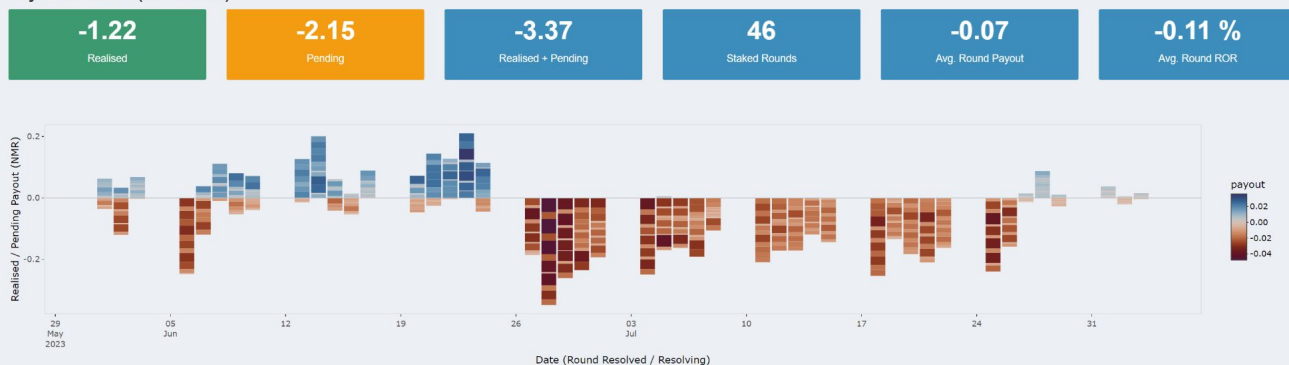
The hedge is not perfect. But it kind of works (so far ...) 🤔

Payouts in NMR (All Models)



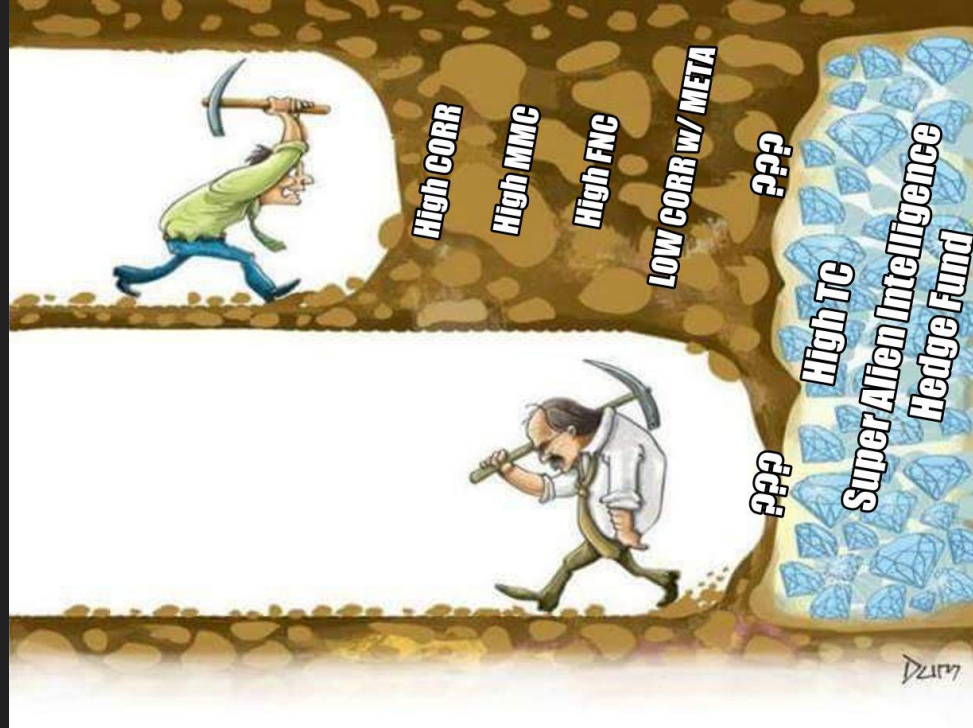
My mainstream +
Anti-mainstream
Models

Payouts in NMR (All Models)

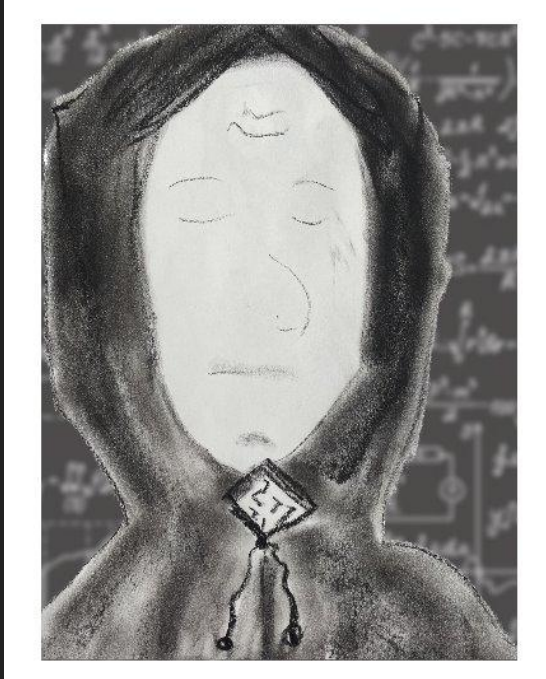


Numerai's
Benchmark
Models

Conclusion - Do Not Give Up



Any Questions?



Contact

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Twitter: @matlabulous